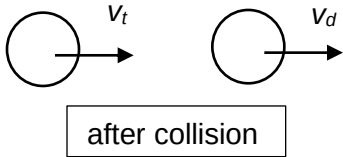


Answers to Prelim Exam H2 Physics Paper 3

1	(a)	The principle of conservation of momentum states that the total momentum of a system of bodies is constant provided <u>no resultant external</u> force acts on the system.	B1
	(b)	The total momentum of the two-nuclei system is non-zero. Thus, if one of the nuclei is at rest, then the other nucleus must possess a non-zero momentum (i.e. it is moving). Hence the two nuclei cannot be at rest simultaneously.	B1 B1
	(c)	Taking right to be positive, By conservation of momentum, $3mv - 2mv = 3mv_t + 2mv_d$ $v = 3v_t + 2v_d$  relative speed of approach = relative speed of separation $2v = v_d - v_t$ Solving simultaneously for v_t and v_d, $v_d = 1.4v$ $v_t = -0.6v$ (i.e. tritium nucleus is travelling leftward)	C1 C1 A1 for both correct answers
		<i>Marker's comments: There were answers that did not treat momentum as a vector. Answers should never be left in fraction form.</i>	
	(d)	Let F_{avg} be the average force exerted by the deuterium nucleus on the tritium nucleus, $F_{avg} = \frac{\Delta p_{tritium}}{t}$ $= \frac{3m(-0.6v - v)}{t}$ $= -\frac{4.8mv}{t} \quad (\text{negative sign indicates it points to the left})$ Magnitude of the average force = $\frac{4.8mv}{t}$	C1 A1
		<i>Marker's comments: The magnitude of the average force should be the same for either deuterium or tritium nucleus. There were mistakes in using the wrong mass or wrong change in velocity. The final numerical value should be shown and not left in fraction form.</i>	